Answer Key & Teacher Notes

Session 1 — Dividing a Fraction by a Fraction · Session 2 — Dividing with Mixed Numbers

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Dividing a Fraction by a Fraction

TALK PROMPTS

- Do you need a common denominator to divide two fractions? Explain your thinking.
- Why does multiplying by the reciprocal give the same answer as dividing?
- Neither number is a whole pound. Why is 7/8 ÷ 1/16 still a whole number?

PARTNER TASK

Honey Jars. One partner writes 3/4 ÷ 1/16 for 3/4 pound of honey and 1/16 pound per jar; the other rewrites it as multiplication and solves (12 jars). Switch roles with Sanding Strips: 11/12 meter torn into 1/12-meter strips (11), and draw the tape diagram.

WATCH FOR

- Flipping both fractions, or flipping the dividend instead of the divisor (3/5 ÷ 1/2 written as 5/3 × 2/1).
- Multiplying straight across without flipping (5/12 ÷ 5/12 written as 25/144).
- Expecting a quotient less than 1 because both numbers are less than 1.

SESSION 2 · TEACHER NOTES

Dividing with Mixed Numbers

TALK PROMPTS

- Why can you not flip a mixed number the same way you flip a fraction?
- After converting, 3 3/4 and 1 1/4 had the same denominator. Did that make the division easier? Say why.

PARTNER TASK

Paint Coats and Yarn Scarves. One partner converts 3 3/4 and 1 1/4 to improper fractions; the other divides (3 coats). Switch roles for 11 1/4 ÷ 2 1/4 (5 scarves), then check each answer by multiplying back.

WATCH FOR

- Converting a mixed number by adding the whole and numerator (4 1/4 written as 5/4).
- Flipping a mixed number piece by piece (1 1/4 flipped to 1 4/1) instead of converting first.
- Dividing whole parts and fraction parts separately (8 3/4 ÷ 1 1/4 as 8 ÷ 1 and 3/4 ÷ 1/4).

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.NS.A.1	1	2	1	1	2	2	3	2	2	1	1
S1 · Form B	6.NS.A.1	1	2	1	1	2	2	3	2	2	1	1
S2 · Form A	6.NS.A.1	1	2	1	1	2	2	3	2	2	1	1
S2 · Form B	6.NS.A.1	1	2	1	1	2	2	3	2	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

reciprocal: A fraction flipped upside down; the two numbers multiply to make **1**.

dividend: The amount you **start** with and split into groups.

divisor: The size of each group; the number you divide **by**.

- To divide by a fraction, multiply by its **reciprocal**.
- Keep the first fraction, change ÷ to ×, and flip only the **second** fraction.
- Check: quotient × divisor = **dividend**.

Example: Write the division: 3/4 ÷ **1/8**. Multiply by the reciprocal: 3/4 × **8** = 24/4. Simplify: 24/4 = **6**. Answer: **6** strips. Check: 6 × 1/8 = 6/8 = 3/4.

YOU TRY

1. a. 3 b. 1/2

2. 9/10 ÷ 3/10 = 9/10 × 10/3 = 3 bags

EXIT TICKET

1. a. 4 b. 6

2. 7/8 ÷ 1/8 = 7/8 × 8 = 7 muffins

3. **A** (3/5 × 7/2)

PRACTICE FORM A

1. a. 5/2 = 2 1/2 b. 3/2 = 1 1/2 c. 4 d. 6 e. 2/5 f. 5/6

2. a. 3/4 ÷ 1/12 = 3/4 × 12 = 36/4 = 9 bowls b. 4/5 ÷ 2/5 = 4/5 × 5/2 = 20/10 = 2 bags

3. 11/12 ÷ 1/12 = 11/12 × 12 = 11 strips

4. 7/8 ÷ 1/16 = 7/8 × 16 = 112/8 = 14 jars

5. Nadia is wrong: she multiplied 5/12 × 5/12 without flipping. 5/12 ÷ 5/12 = 5/12 × 12/5 = 1 piece. A number divided by itself is 1.

6. a. 15/16 − 3/16 = 12/16 = 3/4 L b. 3/4 ÷ 1/16 = 3/4 × 16 = 12 jars

7. a. 9/10 ÷ 3/10 = 3 large projects b. 9/10 ÷ 1/10 = 9 small projects c. 3/10 is 3 times 1/10, so 3 times as many small pieces fit in the same 9/10 meter.

8. a. 5/6 ÷ 1/18 = 5/6 × 18 = 90/6 = 15 name tags b. 1/18 is a very small piece, so many of them fit in 5/6 yard.

9. Only the divisor flips. Keep 3/5, change ÷ to ×, flip 1/2 to 2/1: 3/5 × 2/1 = 6/5 = 1 1/5.

10. **B** (3)

11. **C** (reciprocal)

Self-check word **DIVISOR**

PRACTICE FORM B

1. a. 7/2 = 3 1/2 b. 6/5 = 1 1/5 c. 5 d. 8 e. 3/8 f. 5/3 = 1 2/3

2. a. 5/6 ÷ 1/18 = 5/6 × 18 = 90/6 = 15 bottles b. 6/7 ÷ 2/7 = 6/7 × 7/2 = 42/14 = 3 portions

3. 9/10 ÷ 1/10 = 9/10 × 10 = 9 strips

4. 3/4 ÷ 1/16 = 3/4 × 16 = 48/4 = 12 jars

5. Kofi is wrong: he multiplied 3/8 × 3/8 without flipping. 3/8 ÷ 3/8 = 3/8 × 8/3 = 1 piece. A number divided by itself is 1.

6. a. 11/12 − 3/12 = 8/12 = 2/3 L b. 2/3 ÷ 1/24 = 2/3 × 24 = 16 jars

7. a. 4/5 ÷ 2/5 = 2 large banners b. 4/5 ÷ 1/10 = 8 small banners c. 2/5 = 4/10, which is 4 times 1/10, so 4 times as many small pieces fit in the same 4/5 meter.

8. a. 2/3 ÷ 1/15 = 2/3 × 15 = 30/3 = 10 name tags b. 1/15 is a very small piece, so many of them fit in 2/3 yard.

9. Only the divisor flips. Keep 4/7, change ÷ to ×, flip 2/3 to 3/2: 4/7 × 3/2 = 12/14 = 6/7.

10. **B** (4)

11. **C** (7/3)

Self-check word **INVERSE**

GUIDED NOTES

mixed number: A **whole** number and a fraction written together.

improper fraction: A fraction whose numerator is **greater** than or equal to its denominator.

simplify: Write the same amount using **smaller** numbers.

- Before dividing, write each mixed number as an **improper** fraction.
- Then multiply by the **reciprocal** of the divisor and simplify.
- You cannot flip a mixed number until it is **one** fraction.

Example: Convert: 3 1/2 = **7/2** and 1 3/4 = **7/4**. Multiply by the reciprocal: 7/2 × **4/7** = 28/14. Simplify: 28/14 = **2**. Answer: She can make **2** loaves.

YOU TRY

1. a. 5/2 b. 10/3 × 3/2 = 5

2. 15/4 ÷ 3/4 = 15/4 × 4/3 = 5 servings

EXIT TICKET

1. a. 13/4 b. 5

2. 15/2 ÷ 3/2 = 15/2 × 2/3 = 5 pieces

3. **A** (7/3 × 6/7)

PRACTICE FORM A

1. a. 22/5 = 4 2/5 b. 9/2 = 4 1/2 c. 3 d. 15/2 = 7 1/2 e. 1/2 f. 2/3

2. a. 4 1/2 = 9/2; 9/2 ÷ 3/4 = 9/2 × 4/3 = 36/6 = 6 pieces b. 15/2 ÷ 5/2 = 15/2 × 2/5 = 3 bags

3. 15/4 ÷ 5/4 = 15/4 × 4/5 = 3 coats

4. 45/4 ÷ 9/4 = 45/4 × 4/9 = 5 scarves

5. Tomás is wrong: he found 4 1/2 × 1/2. 4 1/2 ÷ 1/2 = 9/2 × 2 = 9 scoops. Each cup holds 2 half-cup scoops, so there must be more scoops than cups.

6. a. 16 1/2 − 4 1/2 = 12 ft b. 12 ÷ 1 1/2 = 12 × 2/3 = 8 pieces

7. a. 7 1/2 ÷ 1 1/4 = 15/2 × 4/5 = 6 spacings b. 6 markers c. No marker sits at 0, so each spacing ends at exactly one marker.

8. 28/3 ÷ 7/3 = 28/3 × 3/7 = 4 pitchers. Check: 4 × 2 1/3 = 9 1/3.

9. 4 1/4 is 4 × 4 + 1 = 17 fourths, not 5/4 (he added 4 + 1). Correct: 17/4 × 2/1 = 34/4 = 8 1/2.

10. **A** (11/4)

11. **B** (make improper fractions)

Self-check word **FLIPPED**

PRACTICE FORM B

1. a. 4 b. 3 c. 5 d. 40/3 = 13 1/3 e. 1/2 f. 3/5

2. a. 5 1/4 = 21/4; 21/4 ÷ 3/8 = 21/4 × 8/3 = 168/12 = 14 sections b. 35/4 ÷ 7/4 = 35/4 × 4/7 = 5 tubs

3. 20/3 ÷ 5/3 = 20/3 × 3/5 = 4 coats

4. 27/2 ÷ 9/4 = 27/2 × 4/9 = 108/18 = 6 scarves

5. Elena is wrong: she found 5 1/2 × 1/2. 5 1/2 ÷ 1/2 = 11/2 × 2 = 11 scoops. Each cup holds 2 half-cup scoops, so there must be more scoops than cups.

6. a. 14 1/4 − 4 1/4 = 10 ft b. 10 ÷ 1 1/4 = 10 × 4/5 = 8 pieces

7. a. 10 ÷ 2 1/2 = 10 × 2/5 = 4 spacings b. 4 stations c. No station sits at 0, so each spacing ends at exactly one station.

8. 35/4 ÷ 5/4 = 35/4 × 4/5 = 7 pitchers. Check: 7 × 1 1/4 = 8 3/4.

9. 3 2/5 is 3 × 5 + 2 = 17 fifths, not 5/5 (she added 3 + 2). Correct: 17/5 × 5/4 = 17/4 = 4 1/4.

10. **B** (14/3)

11. **C** (2)

Self-check word **INVERTS**

SHEET 1 · DIVIDING A FRACTION BY A FRACTION

1. 4/5 ÷ 1/10 = 8
2. a. 6 b. 3 c. 5/2 = 2 1/2 d. 7/4 = 1 3/4 e. 2/3 f. 2
3. 1/2 ÷ 2/3 = 1/2 × 3/2 = 3/4 of the recipe
4. 5/6 ÷ 5/12 = 5/6 × 12/5 = 60/30 = 2 bookmarks
5. She did not flip the divisor (she multiplied by 2/3 instead of its reciprocal 3/2). Correct: 4/5 × 3/2 = 12/10 = 6/5 = 1 1/5.

SHEET 2 · DIVIDING WITH MIXED NUMBERS

1. a. 6 b. 4 c. 1 1/2 d. 4 1/2 e. 5 f. 3/4
2. a. 17/5 b. 29/6
3. 8 3/4 ÷ 1 1/2 = 35/4 × 2/3 = 35/6 = 5 5/6, so 5 full bowls; 5 × 1 1/2 = 7 1/2, and 8 3/4 − 7 1/2 = 1 1/4 cups left
4. 7 1/2 ÷ 1 1/4 = 15/2 × 4/5 = 60/10 = 6 hours
5. less than 6 1/2 (the divisor is greater than 1; 6 1/2 ÷ 1 1/4 = 5 1/5)
6. **B** (2)
7. **C** (8)

October 2026 conceptual additions — teacher key

Session 1 · Dividing a Fraction by a Fraction

One piece is 2/8 m. Five eighths contain 2 1/2 groups of two eighths. 5/8 ÷ 1/4 = 5/2 = 2.5.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: One piece is 2/8 m. Five eighths contain 2 1/2 groups of two eighths. 5/8 ÷ 1/4 = 5/2 = 2.5.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Dividing with Mixed Numbers

7/2 ÷ 3/4 = 14/3 = 4 2/3. Four complete strips use 3 m, leaving 1/2 m. The fractional quotient is part of a strip-length unit.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: 7/2 ÷ 3/4 = 14/3 = 4 2/3. Four complete strips use 3 m, leaving 1/2 m. The fractional quotient is part of a strip-length unit.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.